PLAYWRIGHT HANDBOOK

1. INTRODUCTION TO PLAYWRIGHT

★ What is Playwright ?

Playwright is an open-source framework by Microsoft for end-to-end testing of modern web applications.

It allows you to automate browsers, intercept network, handle multiple tabs, iframes, and much more.

★ Why Playwright ?

- ✓ Fast and reliable
- ✓ Cross-browser testing (Chromium, Firefox, WebKit)
- ✓ Auto-waiting and built-in assertions
- ✓ Rich set of APIs
- ✓ Supports Multiple Languages
- ✓ Powerful Debugging Tools
- ✓ Handles modern web apps easily (SPA, AJAX, Shadow DOM, etc.)

★ Key Features

- Cross-browser testing
- Auto-waiting
- Headless & Headed mode
- Parallel Execution
- Network Interception
- Screenshots, Videos, Traces
- Mobile Emulation
- Support for Multiple Languages

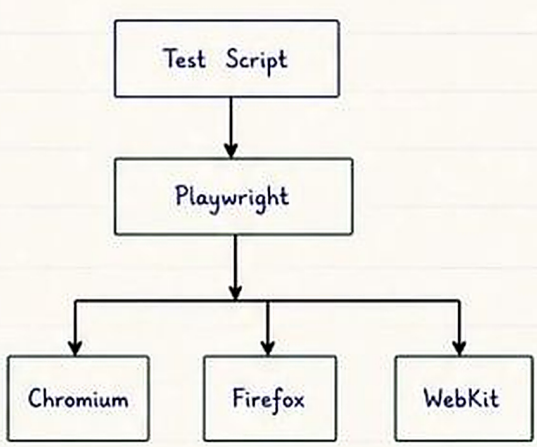
★ Supported Browsers

Chromium	Firefox	WebKit
(Chrome, Edge, Brave, etc.)	(Firefox Browser)	(Safari Browser)

★ Supported Languages

JavaScript	TypeScript	Python	.NET (C#)	Java

★ Playwright Architecture



★ Installation

1. Using npm (Node.js)

```
npm init playwright@latest
        OR
npm install @playwright/test
```

This will install Playwright and all required browsers (Chromium, Firefox, WebKit).

2. Install Browsers

```
npx playwright install
```

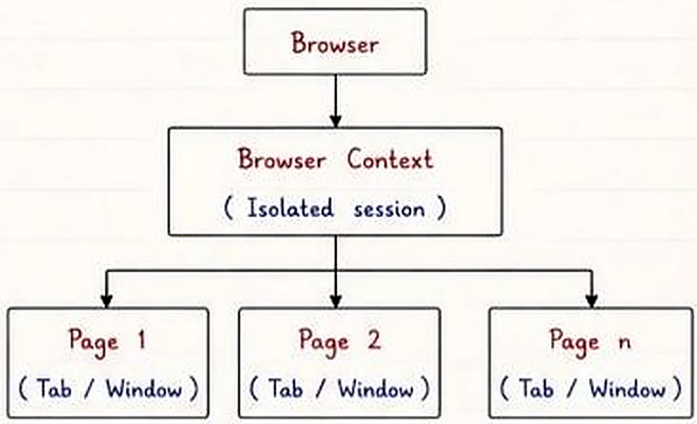
PLAYWRIGHT HANDBOOK

2. BROWSER & PAGE OPERATIONS

★ What is Browser, Context & Page ?

- **Browser** : The actual browser (Chromium, Firefox, WebKit)
- **Context** : An isolated session in a browser. It has its own cookies, storage, cache, etc.
- **Page** : A single tab or window inside a context where we perform actions.

★ Browser → Context → Page



★ Launch Browser

```
const { chromium, firefox, webkit } = require('@playwright/test');

const browser = await chromium.launch({
    headless: false,     // true for headless mode
    slowMo: 50           // slow down (in ms)
});
```

★ Headless vs Headed

Headless (headless: true)	Headed (headless: false)
• Browser runs in background	• Browser window is visible
• Faster execution	• Slower than headless
• No UI is displayed	• Good for debugging
• Ideal for CI/CD	• Used in local execution

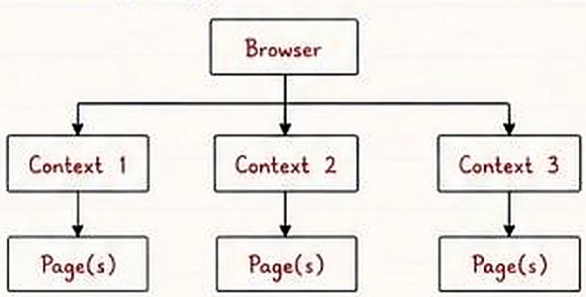
★ Create Context & Page

```
const context = await browser.newContext({
    viewport: { width: 1280, height: 720 }, // set viewport
    ignoreHTTPSErrors: true,
    baseURL: 'https://example.com'          // base url
});

const page = await context.newPage();     // open new tab
await page.goto('/login');                // navigate
```

★ Multiple Contexts

We can create multiple contexts in a single browser for different sessions.



★ Common Page Operations

- `await page.goto('https://example.com');` // navigate
- `await page.title();` // get page title
- `await page.url();` // get current url
- `await page.goBack();` // back
- `await page.goForward();` // forward
- `await page.reload();` // reload
- `await page.close();` // close current tab

★ Close Browser

```
await context.close();     // close context
await browser.close();     // close browser
```

★ Key Points

PLAYWRIGHT HANDBOOK

3. LOCATORS

★ What is a Locator ?

Locator is an object used to find and interact with elements on the page. Playwright provides multiple ways to locate an element.

★ Common Locator Strategies

- Role : Based on ARIA role and name
- Text : Based on visible text
- Label : Based on label associated with input
- Placeholder : Based on placeholder text
- Test Id : Based on data-testid attribute
- CSS : Using CSS selectors
- XPath : Using XPath selectors

★ Locator Methods

Method	Example
locator()	page.locator('css=button')
getByRole()	page.getByRole('button', { name: 'Submit' })
getByText()	page.getByText('Login')
getByLabel()	page.getByLabel('Email')
getByPlaceholder()	page.getByPlaceholder('Enter email')
getByTestId()	page.getByTestId('user-name')

★ Locator Priority (Recommended Order)

getByRole > getByLabel > getByText > getByPlaceholder > getByTestId > CSS / XPath

★ Chaining Locators

We can narrow down element search using chaining.

```
// Example
const row = page.locator('table').locator('tr').nth(1);
const cell = row.locator('td').nth(2);
await cell.click();
```

★ Locator Operators

- locator.first() – first matching element
- locator.last() – last matching element
- locator.nth(i) – element at index i
- locator.filter() – filter locator by condition
- locator.and() – combine two locators
- locator.or() – match either locator

★ CSS vs XPath

CSS Selector	XPath
Faster	Slower
Easier to write	Complex syntax
Most preferred	Useful for complex DOM

★ Best Practices

- ✓ Use getByRole() whenever possible.
- ✓ Avoid CSS/XPath unless necessary.
- ✓ Use data-testid for stable automation.
- ✓ Avoid locating elements using index unless no other option.

PLAYWRIGHT HANDBOOK

4. USER ACTIONS

★ What are User Actions ?

User Actions are the operations we perform on web elements to simulate real user behavior. Playwright provides rich set of actions that are auto-wait enabled and reliable.

★ Action Flow

★ Common User Actions

- Click : Click on an element → click()
- Double Click : Double click on an element → dblclick()
- Fill : Enter text in input field → fill()
- Clear : Clear text from input field → clear()
- Press Keys : Press keyboard keys → press()
- Check : Check checkbox → check()
- Uncheck : Uncheck checkbox → uncheck()
- Select Option : Select from dropdown → selectOption()
- Hover : Hover over an element → hover()
- Drag and Drop: Drag element and drop → dragTo()
- Upload File : Upload file in input → setInputFiles()

★ Example : Fill and Click

```
await page.fill('#username', 'aman');
await page.fill('#password', 'pass123');
await page.click('button[type="submit"]');
```

★ Input Actions

Action	Description
fill()	Fills the element with text.
type()	Types like a user (one by one).
clear()	Clears the element value.
press()	Press a key or combination. (e.g.'Enter')
setInputFiles()	Upload file to input[type="file"]

★ Select & Check Actions

Action	Description
check()	Checks a checkbox.
uncheck()	Unchecks a checkbox.
selectOption()	Select option in dropdown.
isChecked()	Verify checkbox is checked.
isSelected()	Verify option is selected.

★ Mouse Actions

- hover() : Hover over an element
- click() : Click on an element
- dblclick() : Double click
- dragTo() : Drag element and drop on target
- moveTo() : Move mouse to position
- mouse.down() : Press mouse button
- mouse.up() : Release mouse button

Drag and Drop Example

```
const source = page.locator('#drag');
const target = page.locator('#drop');
await source.dragTo(target);
```

★ Key Points

PLAYWRIGHT HANDBOOK

5. ASSERTIONS & WAITING

★ What are Assertions ?

Assertions are used to verify the expected state of elements on the page.
Playwright uses `expect()` API for performing assertions.

★ Basic Assertion Syntax

```
import { expect } from '@playwright/test';

await expect(locator).toBeVisible();
```

Locator — Assertion

★ Common Assertions

Assertion	Description
toBeVisible()	Element is visible
toBeHidden()	Element is hidden
toHaveText(text)	Element has exact text
toContainText(text)	Element contains text
toHaveValue(value)	Input has expected value
toHaveAttribute(name, value)	Element has attribute value
toBeEnabled()	Element is enabled
toBeDisabled()	Element is disabled
toHaveClass(className)	Element has class
toHaveURL(url)	Page URL matches
toHaveTitle(title)	Page title matches

★ Auto Waiting (Built-in)

Playwright automatically waits for element to become ready before performing actions.

Auto-wait Conditions

- Element is attached to DOM
- Element is visible
- Element is stable
- Element is enabled
- Element is not receiving events

★ Explicit Waiting

We can use explicit waits when needed.

- waitForSelector()
- waitForLoadState()
- waitForURL()
- waitForTimeout()
- waitforResponse()
- waitforFunction()

★ Wait Methods

Method	Description	Example
waitForSelector()	Wait for element to appear in DOM	`await page.waitForSelector('#btn');`
waitForLoadState()	Wait for page load state	`await page.waitForLoadState('networkidle');`
waitForURL()	Wait for URL to match	`await page.waitForURL('**/dashboard');`
waitForResponse()	Wait for response matching URL/predicate	`await page.waitForResponse('**/api/data');`
waitForTimeout()	Wait for specified time (ms)	`await page.waitForTimeout(2000);`

★ Best Practices

- ✓ Prefer auto-wait over explicit wait.
- ✓ Use web-first assertions (expect()).
- ✓ Avoid waitForTimeout() unless absolutely necessary.
- ✓ Use specific locators to reduce flakiness.
- ✓ Wait for network responses instead of hard waits.

★ Example : Assertion with Waiting

```
await page.goto('https://example.com');
await page.waitForLoadState('networkidle');  // wait for page load
```

Wait for page

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6. FRAMES, WINDOWS & ALERTS

★ Frames (iframes)

Web pages may contain frames (iframes).
Playwright provides special locators to work with them.

- frameLocator()
- frame()
- contentFrame()

★ Frame Example

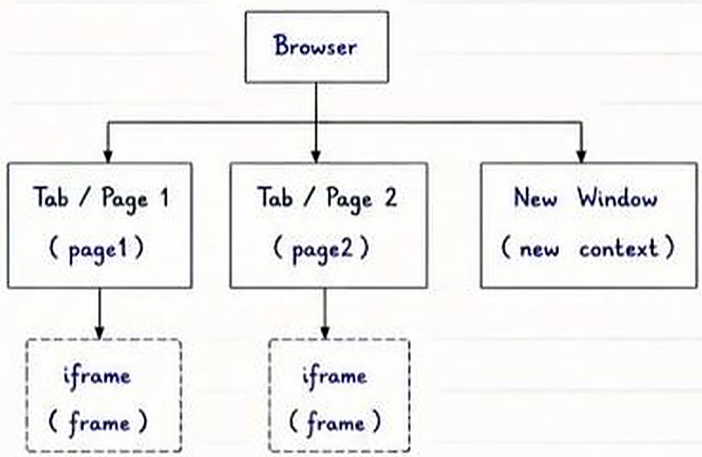
```
const frame = page.frameLocator('#my-frame');
await frame.locator('input[name="username"]').fill('aman');
await frame.locator('button[type="submit"]').click();
```

★ Multiple Windows / Tabs

- Playwright treats each tab/window as a Page.
- We can open new page and switch between them.

```
const page1 = await context.newPage();
await page1.goto('https://example.com');
const [page2] = await Promise.all([
    context.waitForEvent('page'),
    page1.click('a[target="_blank"]')
]);
await page2.goto('https://example.org');
```

★ Browser Window Structure



★ Alerts / Dialogs

Playwright can handle JavaScript dialogs.

- alert() : Information dialog
- confirm() : Confirmation dialog
- prompt() : Input dialog
- dialog event is used to handle them.

★ Alert / Dialog Example

```
page.on('dialog', async dialog => {
    console.log(dialog.type());  // alert, confirm, prompt
    console.log(dialog.message());
    await dialog.accept();       // accept
    // await dialog.dismiss();   // cancel
});
```

★ Dialog Methods

Method	Description
dialog.type()	Returns type (alert, confirm, prompt)
dialog.message()	Returns message text
dialog.accept([text])	Accepts the dialog (For prompt, you can pass text)
dialog.dismiss()	Dismisses the dialog

★ Key Points

- ✓ Use frameLocator() to interact with elements inside iframes.
- ✓ Each tab/window is a separate Page.
- ✓ Use context.waitForEvent('page') to capture new tabs/windows.
- ✓ Always handle dialogs to avoid test failures.

★ In Simple Words

PLAYWRIGHT HANDBOOK

7. API TESTING & NETWORK

★ Why API Testing in Playwright ?

- Playwright has powerful API testing capabilities using APIRequestContext.
- We can test backend APIs, seed data, authenticate and perform CRUD operations.
- Faster and independent of UI.

★ APIRequestContext

It is used to send HTTP requests.

Created using request fixture.

```
test('API test', async ({ request }) => {
  const res = await request.get('https://api.example.com/users');
  console.log(res.status());
});
```

★ HTTP Methods

Method	Description
GET	Retrieve data from server
POST	Create new resource
PUT	Update existing resource
PATCH	Partially update resource
DELETE	Delete resource

★ API Examples

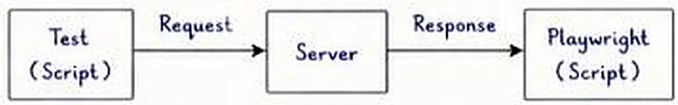
```
// GET Request
const res1 = await request.get('/users');
expect(res1.status()).toBe(200);

// POST Request
const res2 = await request.post('/users', {
  data: { name: 'Aman', email: 'aman@test.com' }
});
expect(res2.status()).toBe(201);

// PUT Request
await request.put('/users/1', { data: { name: 'Aman Saini' } });

// DELETE Request
await request.delete('/users/1');
```

★ Request Options

- headers : Add custom headers
- params : Query parameters
- data : JSON body
- timeout : Set request timeout
- response() : Access response object

★ Network Monitoring

- We can listen to network events during test execution.
- Useful to verify API calls, status codes, payloads etc.



★ Listen to Requests

```
page.on('request', request => {
  console.log('Request URL:', request.url());
  console.log('Request Method:', request.method());
});
```

★ Listen to Responses

```
page.on('response', response => {
  console.log('Response URL:', response.url());
  console.log('Status:', response.status());
});
```

★ Route Interception (Mock / Modify)

- We can intercept and modify network requests.
- Useful for mock APIs, stubbing data etc.

```
await page.route('**/api/data', route => {
  route.fulfill({
    status: 200,
```

★ Key Points

✓ Use API testing to prepare data and validate backend.

✓ API tests are faster and more reliable.

✓ Use route interception to mock or modify APIs.

8. SCREENSHOTS, VIDEOS & TRACE VIEWER

Playwright provides powerful debugging and reporting tools to help us understand test failures easily.

★ Screenshots

- Capture screenshots of page or specific element.
- Helpful for debugging and in reports.

Method	Description
page.screenshot()	Full page screenshot
locator.screenshot()	Element screenshot
fullPage: true	Full scroll screenshot
path	Save screenshot to file

Example :

```
// Full page screenshot
await page.screenshot({ path: 'screens/home.png' });

// Element screenshot
await page.locator('#submit').screenshot({
   path: 'screens/submit.png'
});
```

★ Video Recording

- Record test execution as a video.
- Enabled in context options.

```
const context = await browser.newContext({
   recordVideo: {
      dir: 'videos/',
      size: { width: 1280, height: 720 }
   }
});
```

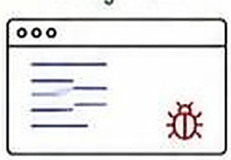
★ Trace Viewer

- Records everything (actions, network, DOM, console).
- Allows step-by-step debugging.
- Very helpful in identifying root cause.

```
// Enable trace
await context.tracing.start({ screenshots: true,
      snapshots: true, sources: true });

// ... run your test ...
await context.tracing.stop({ path: 'trace.zip' });
```

★ Debugging Tools

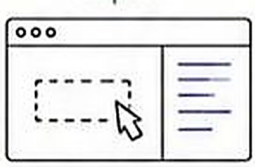
Debug Mode



Run tests in debug mode to pause on failures.

npx playwright test --debug

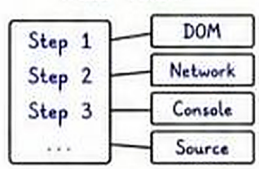
Inspector



Inspect elements and generate locators.

npx playwright codegen

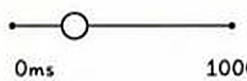
Trace Viewer



Open trace.zip in Trace Viewer to analyze.

npx playwright show-trace trace.zip

SlowMo



Slow down execution to see actions in slow motion.

slowMo: 500

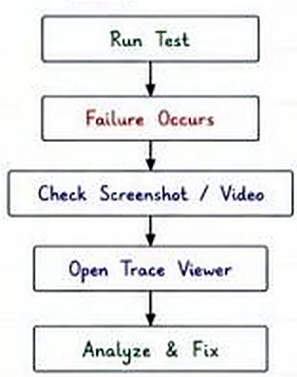
★ Common Screenshot Options

Option	Description
fullPage	Capture full scroll page
clip	Capture specific area
type	png / jpeg
quality	Image quality (0 - 100)
animations: 'disabled'	Disable animations
caret: 'hide'	Hide text caret
scale: 'css'	Use CSS pixels

★ When to Use What ?

Tool	Use Case
Screenshots	Quick visual proof
Videos	See the entire execution
Trace Viewer	Deep debugging
Inspector	Create accurate locators
Debug Mode	Pause and step through

★ Flow



★ Key Points

- ✓ Take screenshots for every important step.
- ✓ Enable video in CI for post execution review.
- ✓ Use inspector to generate stable locators.
- ✓ Debug mode is very helpful during test authoring.

9. CONFIGURATION, REPORTS & PARALLEL EXECUTION

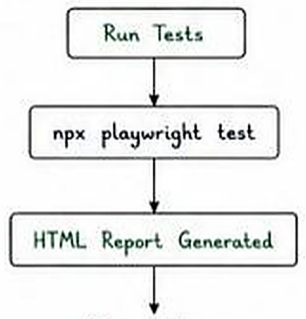
★ Configuration (playwright.config.ts)

All test settings are defined in config file.

```
import { defineConfig } from '@playwright/test';
export default defineConfig({
  testDir: './tests',
  timeout: 30*1000,
  expect: { timeout: 5000 },
  use: {
    headless: true,
    baseURL: 'https://example.com',
    viewport: { width: 1280, height: 720 },
    screenshot: 'only-on-failure',
  },
  projects: [
    { name: 'chromium', use: { browserName: 'chromium' } },
    { name: 'firefox', use: { browserName: 'firefox' } },
    { name: 'webkit', use: { browserName: 'webkit' } },
  ],
});
```

★ Reports

- Playwright generates HTML report.
- Easy to understand with screenshots, videos, logs and steps.

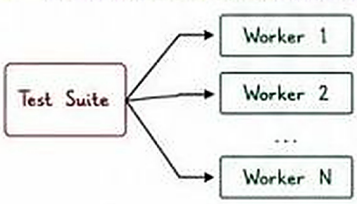


Report Includes

- Test status
- Screenshots
- Videos
- Logs
- Steps
- Duration

★ Parallel Execution

- Tests can run in parallel to save time.
- Use workers to control parallelism.



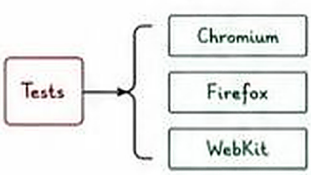
Run in parallel
npx playwright test
--workers=4

★ Workers Example

Workers	Description
1	Run tests one by one (default)
2	Run tests using 2 parallel workers
4	Run tests using 4 parallel workers
100%	Use all available CPU cores

★ Projects

- Run same tests on multiple browsers/devices.
- Configured in playwright.config.ts



Benefits

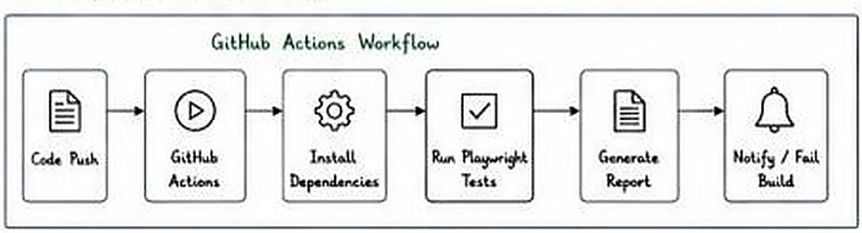
- ✓ Cross-browser testing
- ✓ Consistent results
- ✓ Easy maintenance

★ Useful CLI Commands

Command	Description
npx playwright test	Run all tests
npx playwright test tests/login.spec.ts	Run single test file
npx playwright test --debug	Run in debug mode
npx playwright test --headed	Run with browser UI
npx playwright show-report	Open HTML report
npx playwright codegen	Generate tests using Codegen
npx playwright install	Install browsers

★ CI/CD Integration

- Playwright integrates easily with CI/CD tools.
- Example with GitHub Actions.




```
- name: Install
  run: npm ci

- name: Run Playwright Tests
  run: npx playwright test

- name: Upload Report
  uses: actions/upload-artifact@v4
  with:
    name: playwright-report
    path: playwright-report/
```

★ Key Points

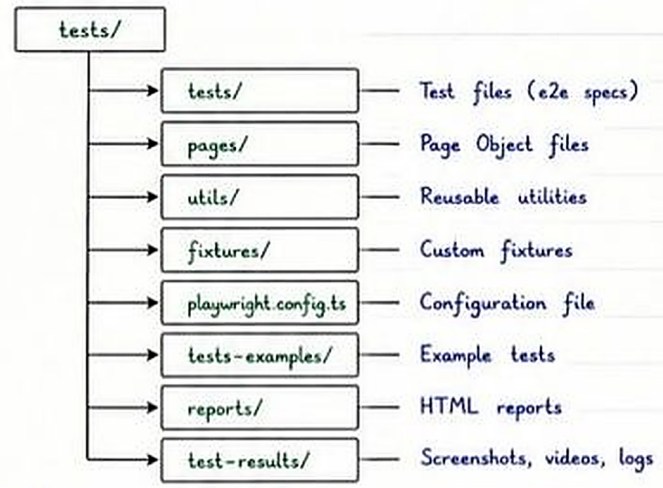
- ✓ Use config file for managing all settings.
- ✓ Run tests in parallel to save execution time.
- ✓ Use multiple projects for cross-browser testing.
- ✓ Integrate with CI/CD for automated test runs.

10. BEST PRACTICES, TIPS & CHEAT SHEET

★ Best Practices

- Use auto-wait and web-first assertions.
- Keep locators stable and reliable.
- Avoid hard waits (waitForTimeout).
- Use Page Object Model for large projects.
- Run tests in parallel for faster feedback.
- Keep tests independent and repeatable.
- Capture screenshots/videos on failure.
- Use meaningful test names and steps.

★ Folder Structure Example



★ Locator Strategies

Strategy	Example
By Role	page.getByRole('button', { name: 'Submit'})
By Text	page.getByText('Submit')
By Label	page.getByLabel('Username')
By Placeholder	page.getByPlaceholder('Enter email')
By Test Id	page.getByTestId('user-input')
By CSS	page.locator('.btn-primary')
By XPath	page.locator('//input[@type="text"]')

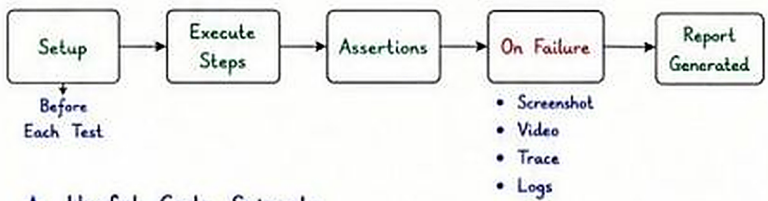
★ Assertions Cheat Sheet

Assertion	What it checks
toBeVisible()	Element is visible
toBeHidden()	Element is hidden
toHaveText()	Element has exact text
toContainText()	Element contains text
toHaveValue()	Input has expected value
toHaveAttribute()	Element has attribute value
toBeEnabled()	Element is enabled
toBeDisabled()	Element is disabled

★ Waits Guide

Wait	Purpose
Auto-wait (default)	Waits for element to be ready
expect() assertions	Retries until condition is met
waitForSelector()	Wait for element in DOM
waitForLoadState()	Wait for page load states
waitForURL()	Wait for URL to match
waitForResponse()	Wait for API response
waitForFunction()	Wait for custom condition

★ Test Flow



★ Command Cheat Sheet

Command	Use
npx playwright test	Run all tests
npx playwright test <file>	Run a specific test file
npx playwright test --ui	Run tests with UI mode
npx playwright test --debug	Run in debug mode
npx playwright show-report	Open HTML report
npx playwright codegen	Generate locators
npx playwright install	Install browsers
npx playwright test --project=chromium	Run on specific browser

★ Useful Code Snippets

```
// Click and assert text
await page.getByRole('button', { name: 'Submit' }).click();
await expect(page.getByText('Success')).toBeVisible();
// Fill input
await page.getByLabel('Email').fill('aman@test.com');
// Select dropdown
await page.getByLabel('Country').selectOption('India');
// Upload file
await page.setInputFiles('input[type="file"]', 'file.pdf');
```

★ Quick Tips

- ✓ Prefer locators like getByRole, getByLabel, getByText.
- ✓ Keep tests small, focused and independent.
- ✓ Use fixture to share setup and data.
- ✓ Always review traces and videos on failure.
- ✓ Run tests on multiple browsers for better coverage.
- ✓ Use parallel execution for faster feedback.

★ In Simple Words

Playwright makes end-to-end testing reliable, fast and developer-friendly with auto-wait,

– Aman Saini